READ SAFETY DIRECTIONS BEFORE OPENING OR USING

Vaniva® 450SC

TYMIRIUM™ technology **Nematicide**

syngenta®

ACTIVE CONSTITUENT: 450 g/L CYCLOBUTRIFLURAM

GROUP	**N-3**	**NEMATICIDE**
GROUP	**7**	**FUNGICIDE**

For the control of root-knot nematodes in Fruiting Vegetables and Cucurbits

Syngenta Australia Pty Ltd
Level 1, 2-4 Lyonpark Road, Macquarie Park NSW 2113

In a transport emergency dial 000, Police or Fire Brigade
For specialist advice in an emergency only, call 1800 033 111 (24 hours)

APVMA Approval no. 91437/132223

SC	Formulation type **Suspension Concentrate**

DIRECTIONS FOR USE

Restraints
DO NOT apply more than one application per crop.
DO NOT apply to dry soil.
DO NOT apply to waterlogged soils.
DO NOT apply via flood irrigation or overhead irrigation systems.
DO NOT apply in tank mixture with other products including fertilisers.
DO NOT apply if heavy rains or storms are forecast within 3 days.
DO NOT irrigate to the point of field runoff for at least 3 days after application.

APPLICATION BY TRICKLE OR DRIP IRRIGATION

Crop	Pest	Rate	Critical Comments
Fruiting vegetables, other than cucurbits (field and protected cropping): including tomato, capsicum, chilli, eggplant	Root-knot nematode (*Meloidogyne* spp.)	550 mL/ha	Apply VANIVA® as a single trickle/drip application within 3 days of transplanting seedlings or planting seeds. Only apply VANIVA® where pre-planting soil counts or paddock history indicate that the root-knot nematode population density is above the economic threshold and significant yield loss is expected. Application placement is important and trickle/drip irrigation tape should be positioned no greater than 12 cm from the plant. VANIVA® application through the trickle irrigation should occur in the second quarter of the irrigation cycle after the soil has been partially wetted up. To limit the potential for development of disease resistance, DO NOT use any fungicide that contains the fungicide chemical class Group 7, SDHI, for the first two foliar fungicide applications in your spray program. Refer to **Mixing and Application** under **General Instructions** for detailed information on application methods. An application of VANIVA® does not provide season long control nor guarantee crops free from nematode root damage at harvest. Best results will be obtained where VANIVA® is used as part of an integrated program aimed at reducing overall nematode populations. **This use may be subject to a CropLife Australia Nematicide and/or Fungicide Resistance Management strategy.**
Fruiting vegetables, cucurbits (field and protected cropping): including cucumber, zucchini, rockmelon, melon, watermelon, pumpkin, Asian melon			

NOT TO BE USED FOR ANY PURPOSE, OR IN ANY MANNER, CONTRARY TO THIS LABEL UNLESS AUTHORISED UNDER APPROPRIATE LEGISLATION

WITHHOLDING PERIODS:
Fruiting Vegetables, Cucurbits: **NOT REQUIRED WHEN USED AS DIRECTED**

EXPORT OF TREATED PRODUCE
Growers should note that maximum residue limits (MRLs) or import tolerances may not exist in all markets for fruiting vegetables and cucurbits treated with VANIVA® 450SC TYMIRIUM technology Nematicide. If you are growing fruiting vegetables and cucurbits for export, please check with Syngenta Australia Pty Ltd for the latest information on MRLs and import tolerances before using VANIVA®.

GENERAL INSTRUCTIONS

VANIVA® will control nematodes present in the treated soil zone. VANIVA will control nematodes that are active in the soil. DO NOT apply VANIVA® when the soil temperature does not allow root-knot nematode activity, generally below 8.5°C.
VANIVA® is effective on all soil types. However, in some soils with high organic matter, control of pest species may be less effective. On heavy soils, care should be taken to ensure a good tilth before application and incorporation.

Mixing
VANIVA® is a Suspension Concentrate (SC) formulation that mixes readily with water and is applied using trickle/drip irrigation. Prior to pouring, shake container vigorously. Prepare the injection solution using a minimum dilution of 550 mL of product in no less than 1 L of water. The volume of the injection solution should not exceed what can be injected in ¼ of the total irrigation volume.

VANIVA® should NOT be mixed with other products including fertilisers.

Application
Drip/trickle irrigation
VANIVA® should be applied through direct injection into drip/trickle irrigation systems. Injection equipment must be accurately calibrated to achieve the required application rates prior to treatment. Even distribution throughout the paddock is essential. Irrigation systems should be monitored periodically to ensure all delivery points are in working order and there are no clogs or blockages. VANIVA® is most effective when drippers are located no greater than 12 cm from the seedling/seed.

The VANIVA® injection solution (see Mixing) should be injected into irrigation systems during the second quarter of the irrigation cycle, after the soil has been partially wetted up. The duration of injection should be long enough to distribute VANIVA® uniformly. This will vary depending on irrigation system setup and design, area of crop treated, emitter type, irrigation flow rates etc. (refer to the manufacturer specifications for the irrigation system). To achieve this, inject VANIVA® over a period of at least twice the duration that it takes irrigation water to travel between the point of injection and the last emitter. This duration should be determined by observing the time it takes for the last emitter to be fully operational after commencement of the irrigation cycle.

The final half of the irrigation cycle is used to flush product from the system and aid in the distribution of VANIVA® into the plant root zone. DO NOT over-irrigate and wash VANIVA® from the root zone. Soil moisture levels should be at or slightly below field capacity at the end of the application and irrigation cycle.

Compatibility
VANIVA® should not be tank mixed with any other products including fertilisers during the application.

Plant back intervals for rotational crops
The following plant back intervals (from the date of treatment) are required for rotational crops:

Crop group	Plant back interval
Cereals, pulses, oil seed crops	15 months
All other crops	None

Resistance Warning

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Although resistance in nematode populations has not been proven, repeated exclusive use of any product may lead to a reduction in control. If further nematicide applications are required rotate using a nematicide with a different mode of action. VANIVA® should be used as part of an Integrated Pest Management (IPM) program to control nematodes. Where possible incorporate IPM programs using cultural practices such as, farm hygiene (including weed control and soil/plant movement), planting of resistant varieties to reduce infestations caused by nematodes, monitoring or other detection methods, proper pest identification and rotation of nematicides with different modes of action will help prevent economic nematode damage.

VANIVA® is also a Group 7, SDHI fungicide. Some naturally occurring individual fungi resistant to VANIVA® and other Group 7 fungicides may exist through normal genetic variability in any fungal population. The resistant individuals can eventually dominate the fungi population if these fungicides are used repeatedly. These resistant fungi will not be controlled by VANIVA® and other Group 7 fungicides, thus resulting in a reduction in efficacy. Since the occurrence of resistant fungi is difficult to detect prior to use, Syngenta Australia Pty Ltd accepts no liability for any losses that may result from the failure of VANIVA® to control resistant fungi.

To limit the potential for development of disease resistance, DO NOT use any fungicide that contains the fungicide chemical class Group 7, SDHI, for the first two foliar fungicide applications in your spray program.

VANIVA® may be subject to specific resistance management strategies. To help prevent the development of resistance to VANIVA®, use VANIVA® in accordance with the current CropLife resistance management strategies. For further information contact your local Syngenta representative, CropLife Australia, farm chemical supplier, local Department of Agriculture or Primary Industries or consultant.

PRECAUTIONS

Re-entry Period: Not required when used as directed.

PROTECTION OF WILDLIFE, FISH, CRUSTACEANS AND ENVIRONMENT

Very toxic to aquatic life. DO NOT contaminate wetlands or watercourses with this product or used containers.

STORAGE AND DISPOSAL

Store in the closed original container in a cool, well-ventilated area. DO NOT store for prolonged periods in direct sunlight.
Triple-rinse containers before disposal. Add rinsings to treatment tank. DO NOT dispose of undiluted chemicals on site. If recycling, replace cap and return clean containers to recycler or designated collection point.
If not recycling, break, crush, or puncture and deliver empty packaging to an approved waste management facility. If an approved waste management facility is not available, bury the empty packaging 500 mm below the surface in a disposal pit specifically marked and set up for this purpose, clear of waterways, desirable vegetation and tree roots, in compliance with relevant local, state or territory government regulations. DO NOT burn empty containers or product.

SAFETY DIRECTIONS

May irritate the eyes. Avoid contact with the eyes.
Wash hands after use.

FIRST AID

First aid is not generally required. If in doubt, contact a Poisons Information Centre (phone 131 126) or a doctor.

SAFETY DATA SHEET

If additional hazard information is required refer to the Safety Data Sheet. For a copy visit our website at www.syngenta.com.au or use the QR code on this label.

DISCLAIMER

This product complies with the specifications in its statutory registration. Implied terms and warranties are excluded. Syngenta's liability for breach of the express or any non-excludable implied warranty is limited to product replacement or purchase price refund. The purchaser must determine suitability for intended purpose and take all proper precautions in the handling, storage and use of the product including those on the label and/or safety data sheet failing which Syngenta shall have no liability.

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